Benchmark: **MA.7.AR.1.1** · Simplify linear expressions Student code: _____ Calculator: no

Learning goal: I can add and subtract linear expressions. (circle your level after scoring)

4	I can write and simplify an expression from a real-world situation.
3	I can add and subtract linear expressions with rational coefficients.
2	I can combine like terms with integer coefficients.
1	I need help to start this benchmark.

1. Write the expression using the fewest possible terms.

$$\left(3+\left(\frac{5}{6}\right)p\right)+\left(\left(\frac{5}{6}\right)p-1\right)$$

2. Which expression is equivalent to the expression shown?

$(0.4x+2)+(1.6x-2.5)$

A. $2x-0.5$

B. $1.5x$

C. $0.64x-0.5$

D. $2x+0.5$

Turn over →

Benchmark: **MA.7.AR.1.1** · Simplify linear expressions Student code: _____

3. What expression makes the equation true?

$$\left(\left(\frac{2}{3}\right)x + 6\right) - (__) = \left(-\frac{3}{8}\right)x - 2$$

4. Write the expression using the fewest possible terms.

$$\left(\left(\frac{4}{3}\right)x - 8\right) - \left(\left(\frac{7}{3}\right)x + 4\right)$$

5. A rectangle has a length of $\left(\left(\frac{5}{2}\right)x + 0.5\right)$ meters and a width of $(x - 2)$ meters.

Write an expression for the perimeter of the rectangle, in meters. Then write it using the fewest possible terms.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.1.1 · MA.7.AR.1.1 · Simplify linear expressions

Unit 9: Algebraic Expressions. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $\left(\frac{5}{3} = 1\frac{2}{3}\right)p + 2$

Group the like terms: $\left(\left(\frac{5}{6}\right)p + \left(\frac{5}{6}\right)p\right) + (3 - 1)$.
p-terms: $\frac{5}{6} + \frac{5}{6} = \frac{5}{3}$, so the p-term is $\left(\frac{5}{3}\right)p$.
Constants: $3 - 1 = 2$.
$= \left(\frac{5}{3}\right)p + 2$

2. Regular **A.** $2x - 0.5$

Add the x-terms: $0.4x + 1.6x = 2x$. Add the constants: $2 - 2.5 = -0.5$.
An x-term and a constant cannot be combined into one term.

3. Regular $\left(\frac{25}{24} = 1\frac{1}{24}\right)x + 8$

The missing expression is the first expression minus the result.
$\left(\left(\frac{2}{3}\right)x + 6\right) - \left(\left(-\frac{3}{8}\right)x - 2\right)$: x-terms $\left(\frac{2}{3}\right)x - \left(-\frac{3}{8}\right)x = \left(\frac{25}{24}\right)x$; constants $6 - (-2) = 8$.
The missing expression is $\left(\frac{25}{24}\right)x + 8$.
Check: $\left(\left(\frac{2}{3}\right)x + 6\right) - \left(\left(\frac{25}{24}\right)x + 8\right) = \left(-\frac{3}{8}\right)x - 2$.

4. Regular $-x - 12$

Subtracting the whole expression changes the sign of each of its terms: $\left(\frac{4}{3}\right)x - 8 - \left(\frac{7}{3}\right)x - 4$.
x-terms: $\left(\frac{4}{3}\right)x - \left(\frac{7}{3}\right)x = -x$. Constants: $-8 - 4 = -12$.
$= -x - 12$

5. Rigor $2\left(\left(\frac{5}{2} = 2\frac{1}{2}\right)x + 0.5\right) + 2(x - 2)$ $\left(\text{or } \left(\left(\frac{5}{2} = 2\frac{1}{2}\right)x + 0.5\right) + (x - 2) + \left(\left(\frac{5}{2} = 2\frac{1}{2}\right)x + 0.5\right) + (x - 2)\right)$**; simplified:** $7x - 3$**. Score: 1 point for a correct perimeter expression and the simplified form** $7x - 3$**; either alone earns 0.**

The perimeter is twice the length plus twice the width: $2\left(\left(\frac{5}{2}\right)x + 0.5\right) + 2(x - 2)$.
Distribute: $2\left(\left(\frac{5}{2}\right)x + 0.5\right) = 5x + 1$ and $2(x - 2) = 2x - 4$.
Combine like terms: $5x + 2x = 7x$, and $1 - 4 = -3$.
The perimeter, in meters, is $7x - 3$.

Re-check strip (3 items, C-AR.1.1-R)

1. Regular $4.93x - 1$

Combine the two x-terms inside the second expression: $0.49x + 0.58x = 1.07x$.
Subtracting the whole expression changes both signs: $6x - 6 - 1.07x + 5$.
x-terms: $6x - 1.07x = 4.93x$. Constants: $-6 + 5 = -1$.
$= 4.93x - 1$

2. Regular **D.** $7.7x + 1.5$

Add the x-terms: $3.2x + 4.5x = 7.7x$. Add the constants: $2 - 0.5 = 1.5$.
An x-term and a constant cannot be combined into one term.

3. Regular $\left(\frac{19}{6} = 3\frac{1}{6}\right)x + 9$

The missing expression is the first expression minus the result.
$\left(\left(\frac{3}{2}\right)x + 8\right) - \left(\left(-\frac{5}{3}\right)x - 1\right)$: x-terms $\left(\frac{3}{2}\right)x - \left(-\frac{5}{3}\right)x = \left(\frac{19}{6}\right)x$; constants $8 - (-1) = 9$.
The missing expression is $\left(\frac{19}{6}\right)x + 9$.
Check: $\left(\left(\frac{3}{2}\right)x + 8\right) - \left(\left(\frac{19}{6}\right)x + 9\right) = \left(-\frac{5}{3}\right)x - 1$.

Comparison strip (2 items, C-AR.1.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $0.97x - 3$

Combine the two x-terms inside the second expression: $0.59x + 0.44x = 1.03x$.
Subtracting the whole expression changes both signs: $2x - 4 - 1.03x + 1$.
x-terms: $2x - 1.03x = 0.97x$. Constants: $-4 + 1 = -3$.
$= 0.97x - 3$

2. Regular $\left(\frac{1}{12}\right)x + 5$

The missing expression is the result minus the first expression.
$\left(\left(\frac{7}{4}\right)x - 2\right) - \left(\left(\frac{5}{3}\right)x - 7\right)$: x-terms $\left(\frac{7}{4}\right)x - \left(\frac{5}{3}\right)x = \left(\frac{1}{12}\right)x$; constants $-2 - (-7) = 5$.
The missing expression is $\left(\frac{1}{12}\right)x + 5$.
Check: $\left(\left(\frac{5}{3}\right)x - 7\right) + \left(\left(\frac{1}{12}\right)x + 5\right) = \left(\frac{7}{4}\right)x - 2$.

Reteach plan — C-AR.1.1

MA.7.AR.1.1 · Simplify linear expressions

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.1.1-R

Benchmark: **MA.7.AR.1.1** · Simplify linear expressions Student code: _____ Calculator: no

1. Write the expression using the fewest possible terms.

$(6x - 6) - (0.49x - 5 + 0.58x)$

2. Which expression is equivalent to the expression shown?

$(3.2x + 2) + (4.5x - 0.5)$

A. $14.4x + 1.5$

B. $7.7x - 1.5$

C. $9.2x$

D. $7.7x + 1.5$

3. What expression makes the equation true?

$\left(\left(\frac{3}{2}\right)x + 8\right) - (__) = \left(-\frac{5}{3}\right)x - 1$

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Re-check (3 items) C-AR.1.1-R

Benchmark: **MA.7.AR.1.1** · Simplify linear expressions Student code: _____ Calculator: no

1. Write the expression using the fewest possible terms.

$(6x - 6) - (0.49x - 5 + 0.58x)$

2. Which expression is equivalent to the expression shown?

$(3.2x + 2) + (4.5x - 0.5)$

A. $14.4x + 1.5$

B. $7.7x - 1.5$

C. $9.2x$

D. $7.7x + 1.5$

3. What expression makes the equation true?

$\left(\left(\frac{3}{2}\right)x + 8\right) - (__) = \left(-\frac{5}{3}\right)x - 1$

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) — C-AR.1.1-X

Benchmark: **MA.7.AR.1.1** · Simplify linear expressions Student code: _____ Calculator: no

1. Write the expression using the fewest possible terms.

$(2x - 4) - (0.59x - 1 + 0.44x)$

2. What expression makes the equation true?

$\left(\left(\frac{5}{3}\right)x - 7\right) + (__) = \left(\frac{7}{4}\right)x - 2$

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) — C-AR.1.1-X

Benchmark: **MA.7.AR.1.1** · Simplify linear expressions Student code: _____ Calculator: no

1. Write the expression using the fewest possible terms.

$(2x - 4) - (0.59x - 1 + 0.44x)$

2. What expression makes the equation true?

$\left(\left(\frac{5}{3}\right)x - 7\right) + (__) = \left(\frac{7}{4}\right)x - 2$

Score: ____ **/ 2** → **copy to your progress strip**